An Interpretable Machine Learning Approach for Dengue Diagnosis Based on Blood Test Patterns

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Abstract

In recent years, over 3.6 million dengue cases and more than 1,900 associated deaths were recorded all over the world. Bangladesh also suffered one of the worst dengue outbreaks in its history. Dengue is often difficult to diagnose early. Although clinical tests such as CBC, NS1 antigen, IgM, and IgG are widely available, interpreting them manually takes more time. This problem is even more challenging in rural healthcare centers. To address this gap, this study introduces an interpretable machine learning model for dengue detection using CBC-based features. The framework includes data preprocessing, removing outliers, and class imbalance handling with SMOTE. The system applies feature selection using Pearson correlation, chi-square, selectKBest, recursive feature elimination, and extra trees. Logistic regression, random forest, and ensemble methods like XGBoost and LightGBM are also used. We tune model hyperparameters with GridSearchCV for classical ML models and Keras Tuner for deep learning models. We also use ensemble methods like voting and stacking, where LightGBM works as the meta-learner. The proposed model achieved an overall accuracy of 95.5%. It shows strong predictive capability in identifying dengue-positive and negative cases. The system is evaluated through 10-fold cross-validation and independent test sets using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix analysis. We use LIME to make the model interpretable and highlight the most important biomarkers. The goal is to provide a reliable, explainable, and low-cost diagnostic tool for both urban and rural healthcare settings.

1. Introduction

The human body has a natural, complex, and active immune system. It protects us from germs, bacteria, and viruses. This system identifies diseases and defends against pathogens. It also prevents new infections. Even with these defenses, bacterial and viral infections remain a big health problem. Today's global issues, like climate change, pollution, population growth, and virus mutation, make the infection more common. Dengue fever is one such viral disease that spreads through mosquito bites. It mainly spreads from Aedes aegypti and Aedes albopictus. Dengue has become a major global health issue. In recent years, around 3.6 million dengue cases and over 1,900 deaths were reported across 94 countries. It shows that dengue patients are increasing day by day worldwide. Tropical and subtropical regions, like Bangladesh, are especially affected. Densely populated cities such as Dhaka often face seasonal dengue outbreaks. Since the first major epidemic in 2000, dengue cases and deaths have continued to rise in Bangladesh. The 2023 outbreak was the most severe in the past 20 years. [1].

Dengue is most common in urban and semi-urban areas with warm and humid climates. Poor sanitation, unplanned city growth, blocked drains, help mosquitoes breed easily. [2]..According to the World Health Organization, dengue fever is most widespread in countries in the Western Pacific, Southeast Asia, and Africa. Between June and October. Bangladesh records the highest number of dengue cases among Southeast Asian countries. In 2023, the number of dengue cases and deaths was higher than in any previous decade. [3]. Therefore, it is important to detect and respond to dengue. Strong prevention plans like mosquito control, public awareness, clean environments, and a better health system need to stop dengue. Otherwise, future outbreaks may cause more cases and deaths. [4].

Dengue infection occurs primarily through the bite of an Aedes mosquito. The virus can stay dormant in the body for a number of days before symptoms appear. Common symptoms include high fever, severe body pain, appetite loss, and skin rashes etc. During the initial one to two weeks of infection, patients may experience a sharp deterioration in health even if the symptoms are not clear. A very low platelet count in blood tests usually indicates a serious condition. Early detection and classification of dengue cases are very important for proper treatment and reducing deaths. Although tests like CBC, NS1 antigen, IgM, and IgG are available. It is still difficult to quickly and accurately predict dengue using

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these reports. Therefore, there is a strong need for an intelligent automated system that can predict dengue infection from clinical test data with high accuracy [5].

Existing research is often limited by small datasets, small models, or does not focus on explainability and real-world use in low-resource healthcare systems. In this study, we propose a unified and interpretable machine learning framework that integrates diverse clinical features to improve dengue prediction accuracy. We also use explainable AI tools like LIME to make sure the predictions are easy for doctors to understand. For this study, we collected a public dataset from a hospital in Dhaka, Bangladesh. The primary accomplishments of this study are as follows:

1. Development of an automated dengue diagnosis framework based on CBC data using advanced machine learning models.
2. Implementation of a comprehensive preprocessing pipeline. It includes median imputation, Z-score–based outlier removal, and SMOTE for class balancing.
3. Application of multiple feature selection techniques such as Pearson correlation, Chi-Square, SelectKBest, Extra Trees, and Recursive Feature Elimination to identify the most critical blood biomarkers.
4. Integration of Explainable AI (LIME) to ensure interpretability and provide a clear visualization of influential diagnostic features in dengue prediction.
5. Construction of a high-performing, interpretable, and clinically deployable model suitable for real-world use in diverse healthcare settings.

2. Literature Review

Recent research on dengue prediction now focuses on using machine learning and clinical data for better early diagnosis and outbreak forecasting. Many studies have looked at different sides of the problem, such as hematological-based detection of dengue patients, using artificial intelligence models to classify infected patients, and predicting outbreaks using climate information. Existing works show that statistical analysis of blood parameters can provide strong clinical indicators of dengue. Traditional machine learning algorithms like Random Forest and SVM have achieved high accuracy when classifying patients using clinical and lab features. Similarly, climate-based models that use temperature, rainfall, and humidity have also worked well for predicting outbreaks in specific regions. Although these studies are useful, they still have several limitations. Most of them rely on data from a single hospital or area. Many do not include deep learning methods or multimodal features. Some models also fail to generalize across different regions and patient groups. To address these gaps, our research aims to develop a complete dengue prediction framework by integrating robust clinical datasets, evaluating multiple machine learning models, and focusing on improving generalization and real clinical usefulness.

Wang et al. (2023) [6] proposed an artificial intelligence which is explainable (XAI) framework for predicting dengue outbreaks, which is combined with machine learning models with an interpretable algorithm to support making better decisions in public health. The aim of this study is to address the problem of understanding the influence of aerial and environmental factors on dengue transmission and ensuring the transparency in prediction models. The datasets that have been worked with were taken from many dengue endemic regions in all weather conditions. The author trained the models, including Random Forest, XGBoost, and LightGBM. These models achieved high predictive performance, and the model that achieved the highest accuracy is the XGBoost model. The main strength of the study was adding the explainability tools to get the contribution of each feature, getting higher model trust, and practical applicability for policymakers. However, there are limitations which include dependence on data availability, and there can be regional bias due to uneven distribution of dengue case data across areas.

Davi et al.[7] proposed a machine learning approach utilizing human genome data to predict dengue flavivirus. By analyzing genetic markers from 102 patients, the study aimed to identify individuals at high risk of severe dengue, even when presenting mild symptoms. Utilizing SVM-RFE for feature selection, their Artificial Neural Network (ANN) model demonstrated superior performance compared to other classifiers, achieving 86% accuracy, 98% sensitivity, and 51% specificity. In the context of recent outbreaks in Bangladesh, Sarma et al. developed an automated dengue prediction model utilizing data from the country's two most populous cities, Dhaka and Chittagong. Their comparative analysis of various machine learning algorithms revealed that the Decision Tree model yielded the most effective results, attaining an accuracy of 79%.[8]

Chaw et al.[9] developed an AI-based automated framework to predict shock development in dengue patients, utilizing physiological data from the University of Malaya Medical Centre. Among the evaluated machine learning models, the Adaptive Logistic Regression yielded the highest F1-score of 0.93, AUC of 0.65, and precision of 86%.

Prome et al. (2024)[10] developed an explainable machine learning model for predicting dengue outbreaks across 11 districts of Bangladesh by analyzing climatic and aerial data. Addressing the problem of early dengue detection using accessible weather parameters to assist in outbreak prevention and public health response is the main focus of this study. The author collected a dataset that contains 1397 samples from January 2011 to July 2021 from the Bangladesh Meteorological and Health Service Departments. After preprocessing the data, 7 machine learning algorithms were applied, with support vector regression showing the best performance. The optimized SVR model got the lowest Mean Absolute Error of 3.865 and Root Mean Squared of 16.04, performing better than other models. They used the Shapash explainable AI framework to interpret the feature importance, and they found that humidity and temperature are the main predictors. The study's main strength is the integration of explainable AI for model transparency and the long-term dataset. However, there are limitations that include limited environmental features and geographic scope, and this limitation can restrict the generalization to larger or future climatic variations.

Fernàndez et al. [11] implemented a logistic regression model for dengue diagnosis, analyzing clinical features from a cohort of 548 patients presenting with febrile illness. The model yielded a diagnostic accuracy of 69.2%, along with a sensitivity of 86.2% and specificity of 27.2%. Islam et al. (2023) did a comparative evaluation of different machine learning algorithms to predict infectious diseases; the focus point was on COVID-19 and dengue to identify the most effective approach. Analyzing the performance differences between regression and ensemble-based models to capture the temporal trends and outbreak patterns from historical data. The author used open datasets of daily reported cases for both diseases. The author applied various models like Linear Regression, Random Forest, Gradient Boosting ad LSTM networks. Among the applied models, the LSTM model showed superior performance with higher prediction accuracy and lower error metrics. It handled the nonlinear temporal dependencies perfectly. The results clearly showed that traditional machine learning techniques perform less than deep learning models in time series prediction. The comprehensive comparison across more than one model and diseases, providing insights into generalizable forecasting strategies, is the main strength of this study. However, this study relied primarily on case count data without incorporating environmental variables, which reduces the models' ability to capture real-world scenarios. This is the limitation of this study.[12].

Mayrose et al. [13] presented an automated prediction framework using blood smear images. It focused on lymphocyte nucleus and platelet features. Using various machine learning techniques, their support vector machine (SVM) classifier achieved superior performance, achieving 95.74% accuracy, 98.15% sensitivity, 92.50% specificity, and 96.36% F1-score. Ong et al. [14] analyzed dengue transmission rates by integrating meteorological information and vector indices across various machine learning algorithms. Using the XGBoost classifier and Boruta feature selection technique, the study achieved 81% accuracy and an AUC of 0.815.

Yang et al. (2023) [15] conducted a multicenter hospital-based study in Dhaka, Bangladesh. It identified demographic, medical, and biochemical vulnerabilities associated with severe dengue and developed an early warning model for clinical use. The study aimed to analyze the combined effects of these variables using statistical and machine learning methods, specifically logistic regression, classification, and regression tree, and random forest models. Data were collected from 1,090 confirmed dengue patients from eight hospitals during the 2019 outbreak through structured questionnaires. The models revealed that respiratory distress, plasma leakage, and hemorrhage were the main independent vulnerabilities. This reached a probability of severe dengue of 92.9% in patients aged 12.5 years who had both respiratory distress and plasma leakage. The random forest model achieved 86.5% accuracy, identifying age as the most influential predictor, followed by education, plasma leakage, platelet count, and respiratory distress. The strengths of this study lie in its large, multicenter data collection and the integration of an advanced machine learning model to capture complex interactions. However, it was limited by its cross-sectional design, single-time biomarker measurements, potential recall bias, and lack of IgG or serotype data to distinguish primary and secondary infections.

Mello-Román et al. [16] developed predictive models for dengue diagnosis utilizing real-world patient data from healthcare centers in Paraguay. Their implemented Artificial Neural Network (ANN) demonstrated robust performance, achieving a classification accuracy of 96%, with sensitivity and specificity reaching 96% and 97%, respectively.

An extensive overview of the existing literature shows that there has been notable progress in automated dengue prediction using traditional machine learning and deep learning models, although most of them have critical gaps.

Table 1
Comparative analysis of dengue prediction techniques and models.

Papers	Motivation	Highlights	Inference	Limitations
[6]	Address environmental factors and ensure transparency.	Proposed explainable AI (XAI) framework using XGBoost.	High model trust and practical applicability for policymakers.	Regional bias due to uneven data distribution.
[7]	Identify high-risk patients using human genome data.	Utilized SVM-RFE for feature selection and ANN.	High sensitivity (98%) in predicting severe dengue.	Low specificity (51%) and limited sample size.
[8]	Predict outbreaks in Dhaka and Chittagong, Bangladesh.	Automated model comparing various ML algorithms.	Decision Tree was most effective with 79% accuracy.	Geographic scope limited to only two cities.
[9]	Predict shock development in patients.	AI framework using physiological clinical data.	Adaptive Logistic Regression yielded F1-score of 0.93.	Low AUC (0.65) suggests issues with class separation.
[10]	Early detection using accessible weather parameters.	Integrated Shapash XAI framework with optimized SVR.	Humidity and temperature identified as main predictors.	Limited geographic scope and environmental features.
[11]	Diagnose dengue from general febrile illness.	Logistic regression model analyzing clinical features.	Strong sensitivity (86.2%) for clinical diagnosis.	Very low specificity (27.2%) and lower accuracy.
[12]	Compare temporal trends of COVID-19 and dengue.	Evaluated traditional ML vs. Deep Learning (LSTM).	LSTM outperformed ML in capturing nonlinear trends.	Relied on case counts only; no environmental variables.
[13]	Automated prediction using blood smear images.	Focused on lymphocyte and platelet features via SVM.	Achieved superior performance with 95.74% accuracy.	Dependence on high-quality imaging.
[14]	Analyze transmission via weather and vector indices.	XGBoost classifier with Boruta feature selection.	Balanced performance with 81% accuracy and 0.815 AUC.	Dependent on availability of vector indices.
[15]	Early warning model for clinical hospital use.	Multicenter study identifying medical vulnerabilities.	Age and respiratory distress are key predictors.	Cross-sectional design; lacks serotype data.
[16]	Diagnosis using real-world healthcare data.	Artificial Neural Network (ANN) for robust classification.	High accuracy (96%) and balanced sensitivity.	May not generalize to different clinical protocols.

As summarized in Table 1, most studies are done on isolated data modalities such as purely meteorological indices or specific clinical parameters, without fully taking into consideration the complex interaction of clinical and environmental factors. Moreover, there is a gap in the existing models and the ones needed to be developed, because only a small minority of such works use state-of-the-art explainable AI (XAI) techniques, and the models that do are typically only a handful. This study will fill these gaps to enhance the reliability of prediction as well as the clinical utility by integrating strong clinical datasets into an elucidable framework.

3. Methodology

In this study, we've used ML to detect dengue-infected patients from CBC tests. The tests for dengue detection, like NS1 antigen tests, RT-PCR, etc, are costly and also they're limited in low-resourced areas [17]. But CBC tests are already standard in hospitals, and low-cost [18]. Our aim is to find out the best machine learning model that can accurately detect dengue-infected patients by only using their CBC data.

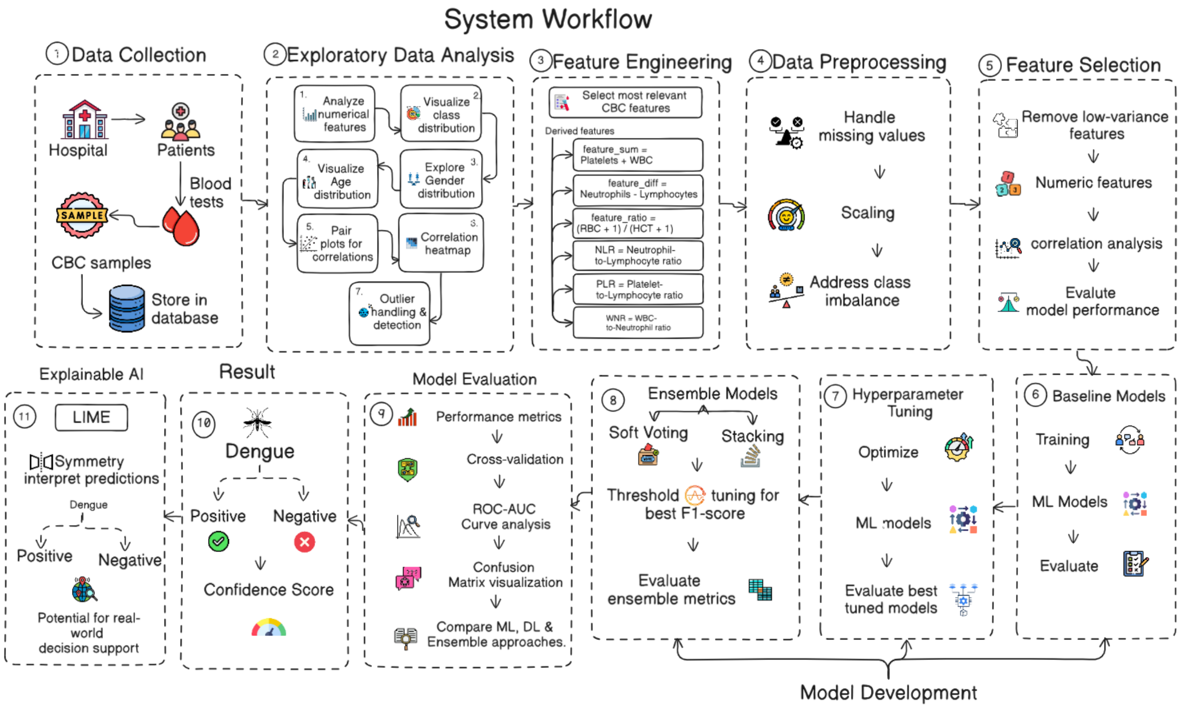
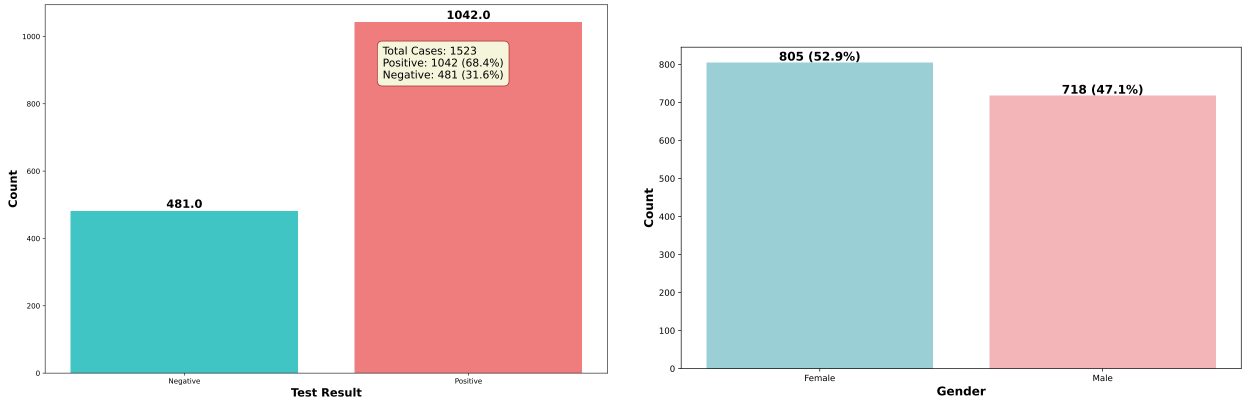


Figure 1: System Workflow of our proposed interpretable ML Approach for Dengue Diagnosis Based on Blood Test Patterns

3.1. Data Collection

A major contribution of this study is the utilization of a publicly available Dengue Fever Hematological Dataset. Data were collected from dengue patients from hospitals in Jamalpur, Bangladesh[19]. The dataset contains 1523 patient samples and 19 hematological features derived from CBC test reports. This dataset provides a broader and more diverse representation of dengue and non-dengue cases. The attributes in the dataset include demographic features such as age and gender. It also has the hematological features like Hemoglobin, Hematocrit (HCT), RBC, WBC, Platelet Count, Neutrophils, Lymphocytes, Monocytes, Eosinophils, Mean Corpuscular Volume (MCV), Mean Corpuscular Hemoglobin (MCH), Mean Corpuscular Hemoglobin Concentration (MCHC), Red Cell Distribution Width-Coefficient of Variation (RDW-CV), Mean Platelet Volume (MPV), Platelet Distribution Width, Platelet Distribution Width (PDW), and PCT. The result column is set as the target label. It indicates whether the patient was diagnosed as dengue positive or negative. It includes both male and female records in the sample distribution. Since the dataset was created from CBC test reports, all the attributes are in standard clinical units. Especially, hematological variations such as low platelet count, decreased WBC count, and hematocrit changes are strongly reflected. They are known diagnostic indicators of dengue fever. This dataset provides a strong foundation for studying the relationships between blood test features and building machine learning models for early and explainable dengue detection. Categorical variables, such as gender, were encoded as Male = 1, Female = 0. Age values were grouped into 5 categories and then one-hot encoded, using 0-10 age group as the reference class. The dataset was then divided into a training set (80%) and a testing set (20%). Stratified splitting was used to keep the same class distribution in both sets.



(a) Distribution of Dengue Classes

(b) Gender Distribution

Figure 2: Overall visualization of Distribution of Dengue Classes (a) & Gender Distribution (b)

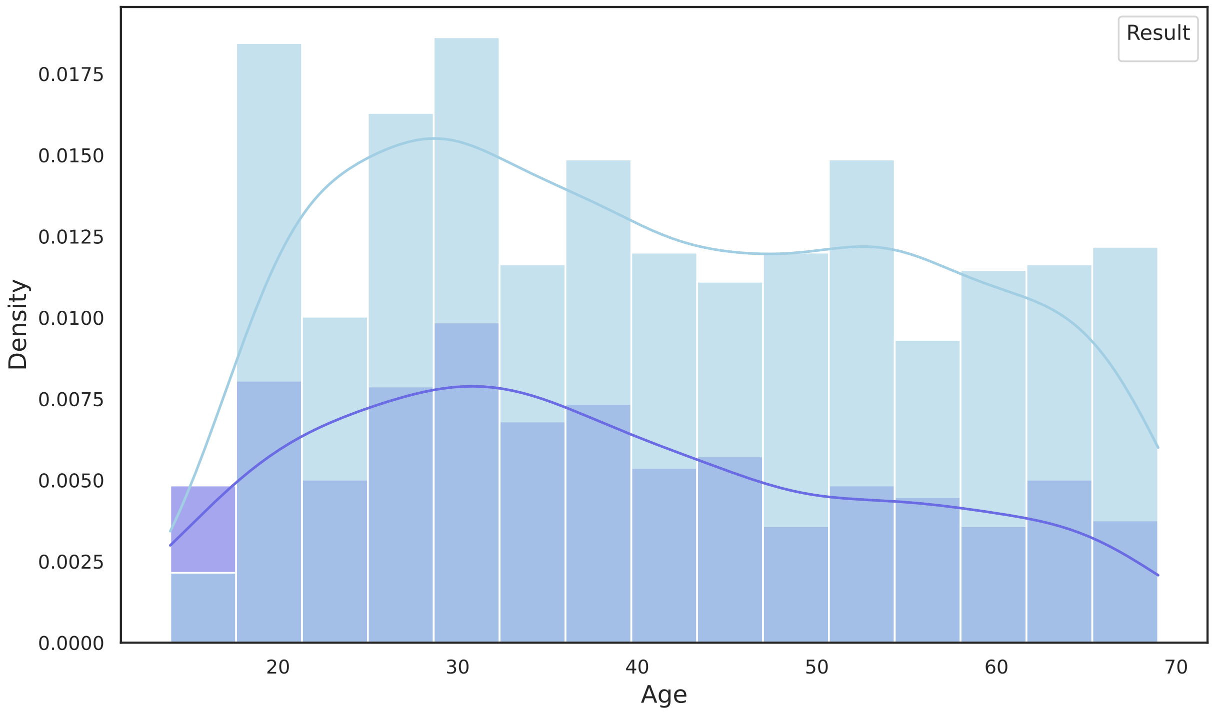


Figure 3: Age Distribution

3.2. Data Preprocessing

We collected and cleaned the data, then implemented a step-by-step preprocessing pipeline to ensure the dataset is suitable for developing robust machine learning and deep learning models. By applying this process, all samples remain consistent, trustworthy, and comparable. It includes handling missing values, encoding categorical features, scaling numerical variables, identifying and removing outliers, addressing class imbalance, and preparing well-balanced training and testing datasets.

	count	mean	std	min	25%	50%	75%	max
Gender	1523	0.4714379514	0.4993475052	0	0	0	1	1
Age	1523	40.31976362	15.16651499	14	28	39	53	69
Hemoglobin	1523	14.51726106	1.61663124	11.4	13.2	14.6	15.8	17.4
Neutrophil	1523	44.43663821	4.196913608	30	42	45	48	52
Lymphocyte	1523	43.53578464	4.930715747	29	41	44	47	50
Monocyte	1523	3.267235719	1.195214687	2	2	3	4	8
Eosinophile	1523	3.076165436	0.9703775517	2	2	3	4	6
RBC	1523	4.765594222	0.7228743987	4	4	5	5	7
HCT	1523	45.19380387	3.551835506	38.1664	42.595	45.55	47.84	51.7512
MCV	1523	89.81424819	5.509963548	80.1	85.1	89.7	94.2	99.6
MCH	1523	29.90773079	1.992889979	26	28.4	30	31.4	33.8
MCHC	1523	32.05488299	1.536648699	28.4128	30.9	32	33.2	34.9
RDW_CV	1523	13.62957781	1.411644661	11.022	12.4	13.8	14.7	16.41
Platelets	1523	173184.2007	64706.3977	78000	114907	167385	228649.5	297710.6
MPV	1523	9.589420223	0.8456271553	8.01	8.9	9.66	10.283	10.99
PDW	1523	15.31817925	1.389389826	12.5	14.23	15.26	16.4	17.93
PCT	1523	0.134307111	0.05004409122	0.002	.106	0.141	0.173	0.21
WBC	1523	5446.518056	942.8505163	4000	4626	5397	6131.5	8000
Result	1523	0.6841759685	0.4649958959	0	0	1	1	1

Table 2
Summary Of Numerical Features

1. **Missing Value Handling**
There were some missing values in hematological features. Those were sealed by using the median value of each feature. The median values were picked because it is more steady and less controlled by outliers than the mean. This proposition safeguarded the realistic figure of the data. By performing this process, not a single sample was removed from the data, which prevented data loss and kept the dataset compatible and authentic.[20, 21]

2. **Encoding Categorical Variables**
Categorical attributes were reshaped into a numerical structure. In the dataset, gender was encoded as Male = one (1) and Female = zero (0). Parallelly, the expected dengue diagnosis output is encoded as Negative = zero (0) and Positive = one (1). This acknowledges machine learning algorithms to illuminate categorical attributes without altering their intrinsic meaning.[22]

3. **Feature Scaling**
All the numerical features were normalized by using the StandardScaler and transforming values so that they have a mean of zero and a standard deviation of one. The scaler was ideal only on the training data, and then used in the test data. This prohibited the data leakage and ensured consistent scaling across training, testing, and deployment stages as well.[20, 23]

4. **Outlier Detection and Removal**
Outliers were detected by using the Z-score method with the numerical blood parameters:

$$Z = \frac{x - \mu}{\sigma} \tag{1}$$

where x is the observed value, μ is the mean, and σ is the standard deviation. Values with $|Z| > 3$ were identified as outliers based on the three-sigma rule. To reduce their effect, extreme values were capped within the 1st–99th percentile range so that we can preserve the statistical integrity of the dataset.[24]

5. **Class Imbalance Handling**
The dataset contained misproportioned data between dengue-positive and negative samples. To precisely address this misproportion, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training

dataset only and ensured balanced class representation while preventing the information leakage into the test dataset. Although the dataset exhibits moderate class imbalance (68.4% positive vs 31.6% negative), SMOTE was applied to improve the model's ability to correctly identify dengue-negative cases.

In medical diagnosis, minimizing false negatives is critical; therefore, enhancing minority class representation improves recall and overall model reliability.

Additionally, class-weighting strategies were also considered. However, SMOTE provided better balance between precision and recall.

A comparative analysis of model performance with and without SMOTE is presented in Section 4.

Model	Acc.	Prec.	Rec.	Specificity	F1	AUC
RF with SMOTE	0.7836065574	0.7848605578	0.9425837321	0.4375	0.8565217391	0.6945524322
RF without SMOTE	0.7573770492	0.7566539924	0.95215311	0.3333333333	0.843220339	0.7027761164
RF with Class-Weight	0.7803278689	0.7730769231	0.961722488	0.3854166667	0.8571428571	0.6806469298

Table 3
Smote Comparison1

Model	Acc.	Prec.	Rec.	Spec.	F1	AUC	Bf_Cl_1	Bf_Cl_0	Af_Cl_1	Af_Cl_0	Imb_Ratio
DATA SUMMARY							833	385	833	666	0.4622
RF with SMOTE	0.7836065574	0.7848605578	0.9425837321	0.4375	0.8565217391	0.6945524322					
RF without SMOTE	0.7573770492	0.7566539924	0.95215311	0.3333333333	0.843220339	0.7027761164					
RF with Class Weight	0.7803278689	0.7730769231	0.961722488	0.3854166667	0.8571428571	0.6806469298					

Table 4
Smote comparison2

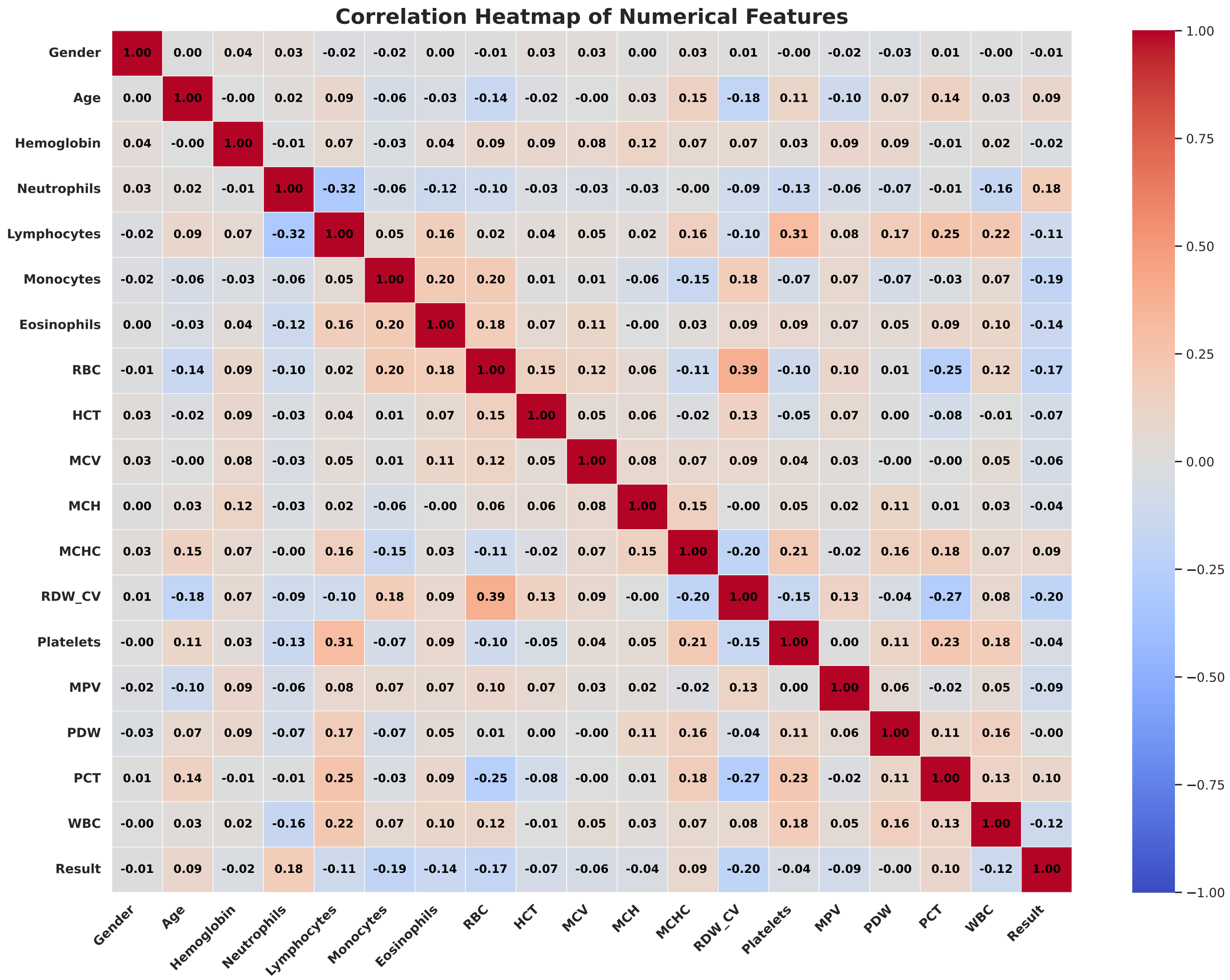


Figure 4: Correlation Heatmap of Numerical Features

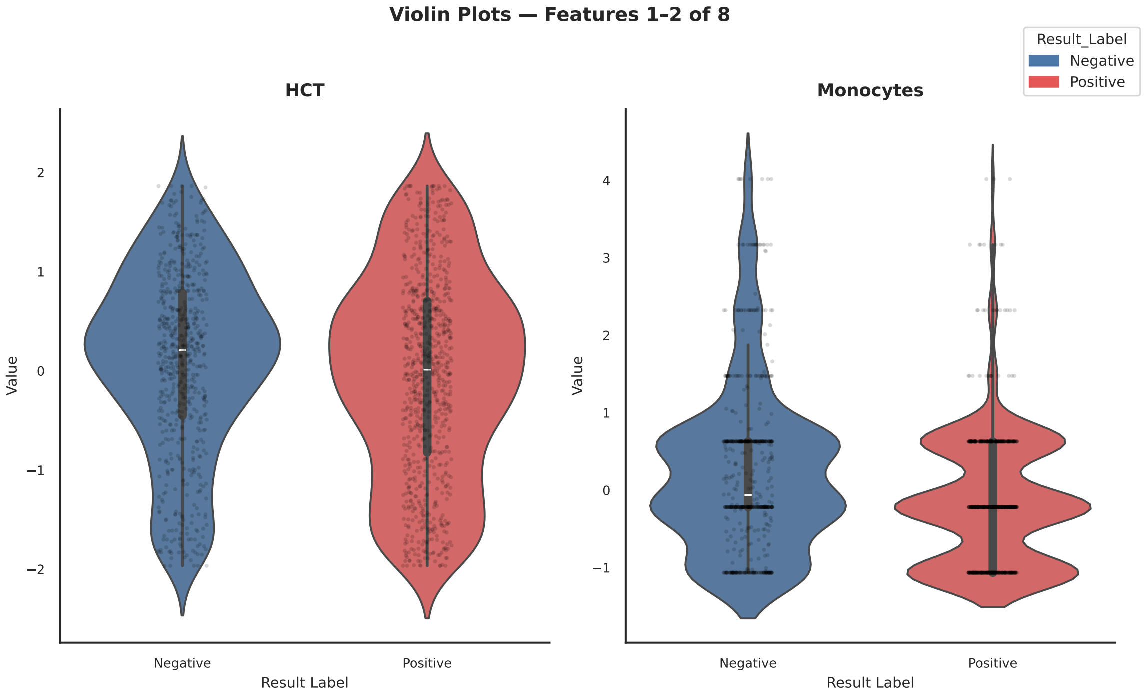


Figure 5: Violin Plots

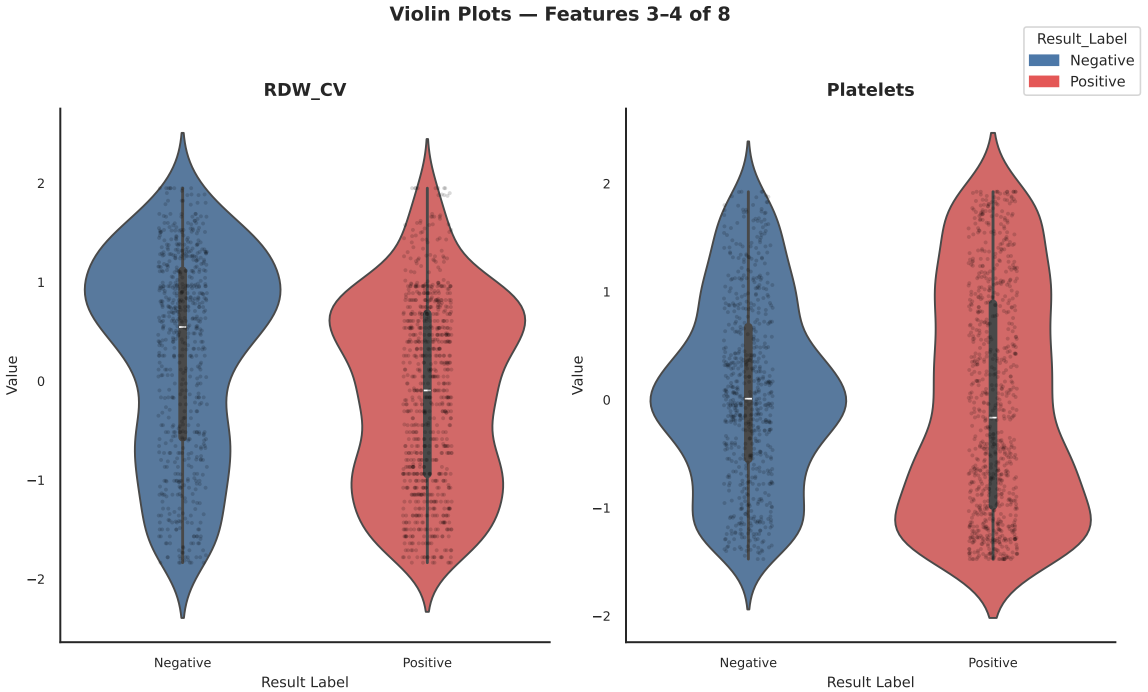


Figure 6: Violin Plots

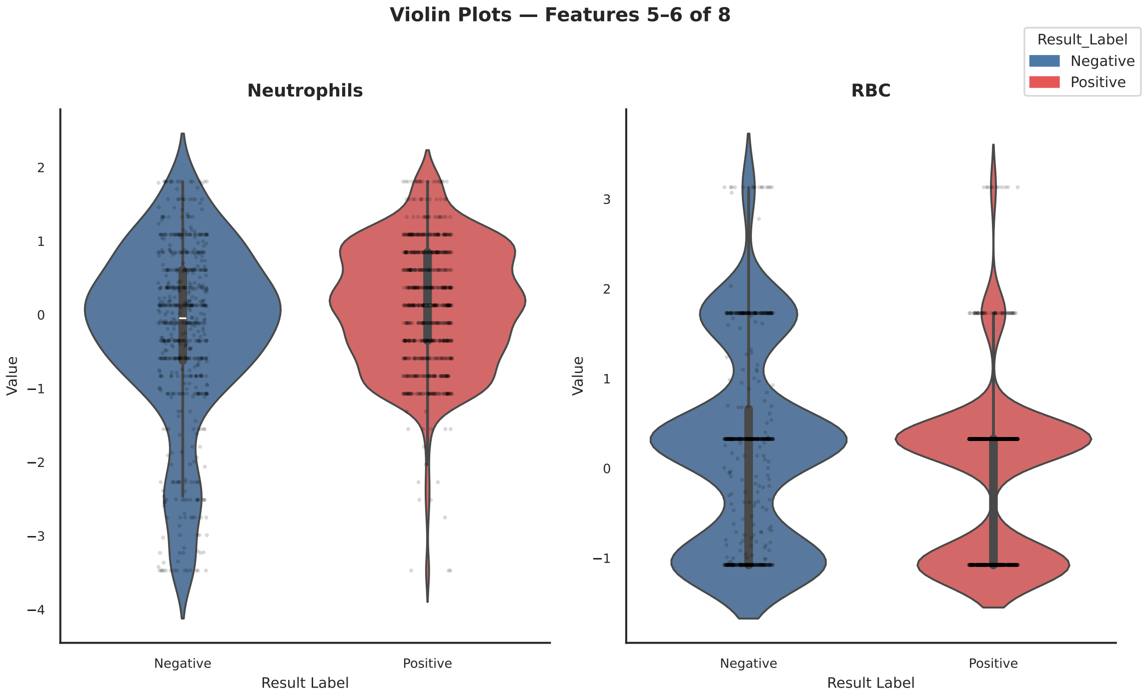


Figure 7: Violin Plots

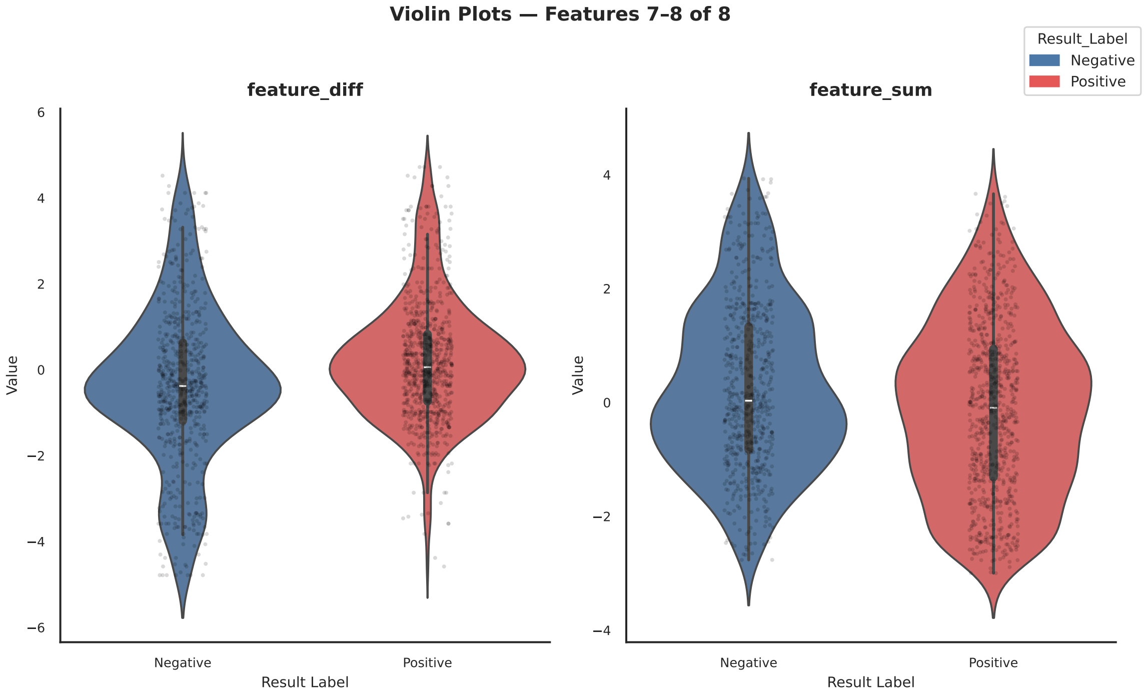


Figure 8: Violin Plots

3.3. Exploratory Data Analysis

Before building machine learning models Exploratory Data Analysis was performed to achieve a deeper understanding of the dataset. At first, the class distribution was examined to check for any imbalance between dengue-positive and negative samples. Then the gender distribution was reviewed to find out the quantity of male and female patients in the dataset. To determine the variation of dengue cases across different age groups, the age distribution was reviewed. To visualize the connection between various blood test features, pair plots were created. On the other hand, a correlation heatmap was created to identify strongly related parameters. In this way multicollinearity and highlighted features were detected that offered distinct and meaningful information. In addition, violin plots for important blood indicators such as WBC count and platelet count were produced to demonstrate how these values differed between dengue-positive and negative groups. One the whole, the EDA outcomes revealed major patterns in blood test characteristics and assisted the feature selection process for the proposed prediction model.

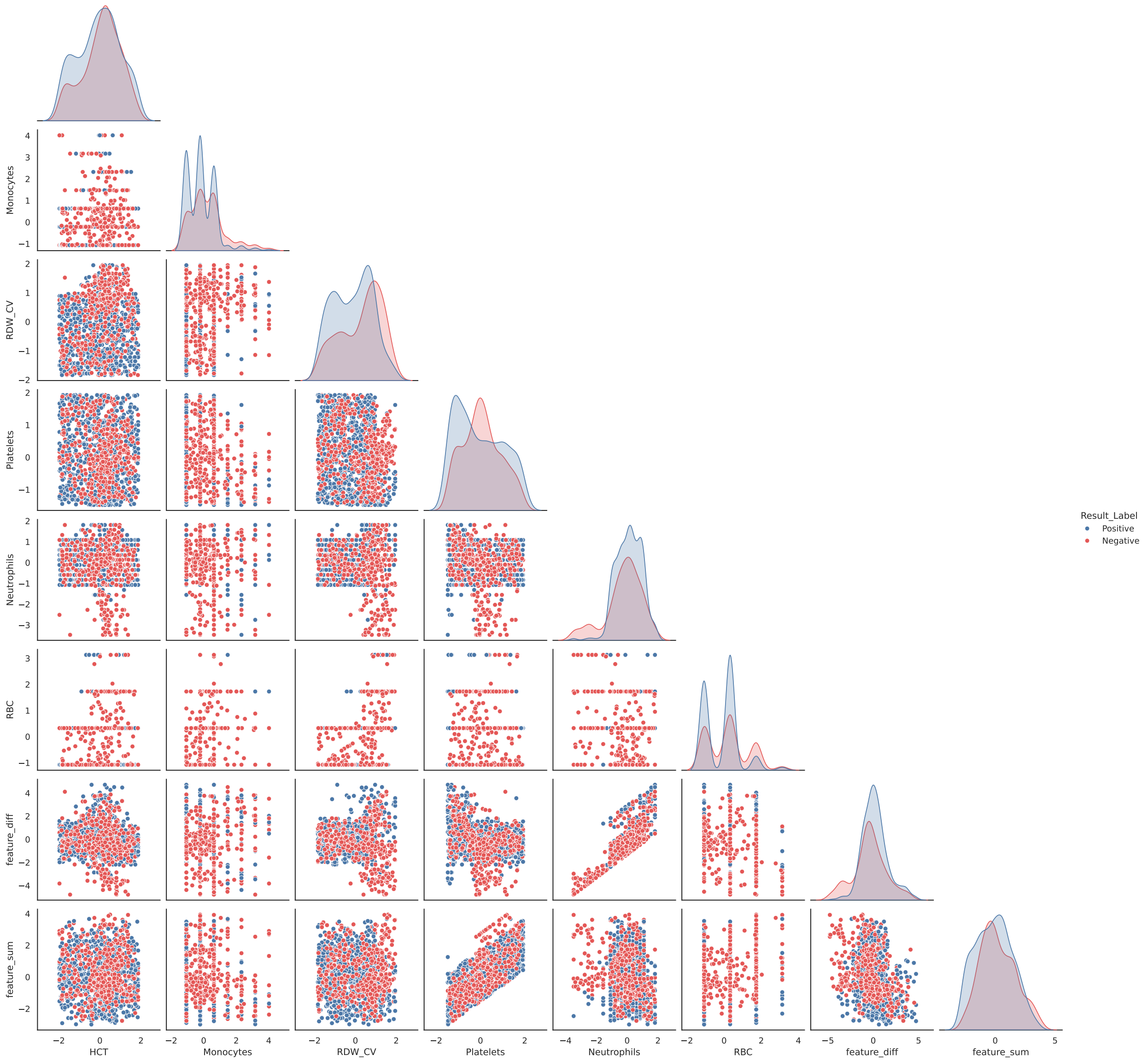


Figure 9: Violin Plots

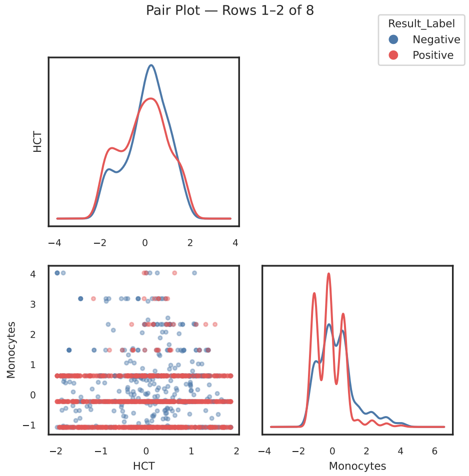


Figure 10: Violin Plots

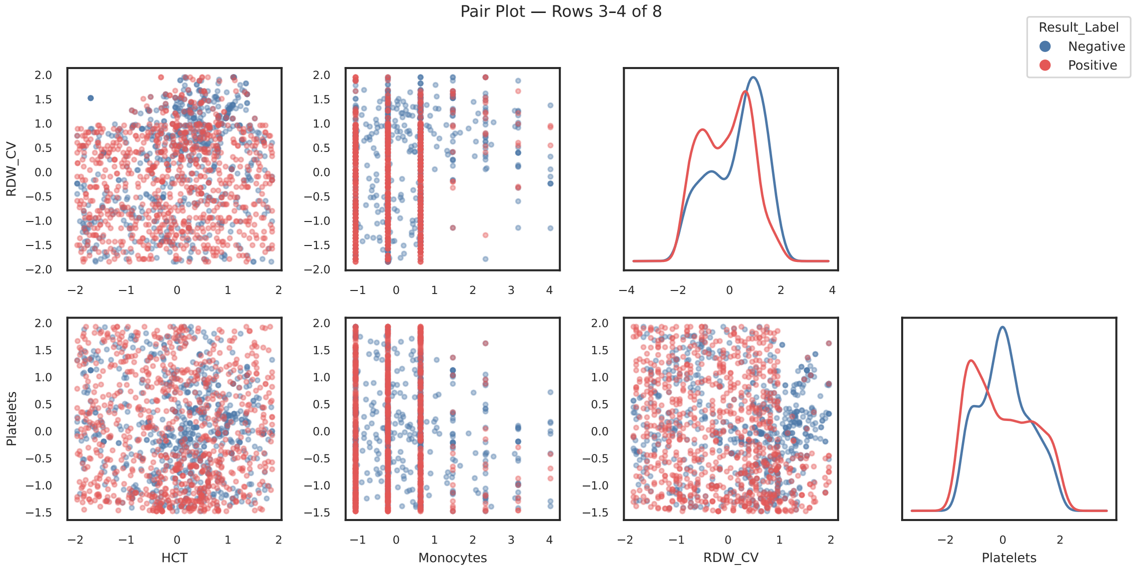


Figure 11: Violin Plots

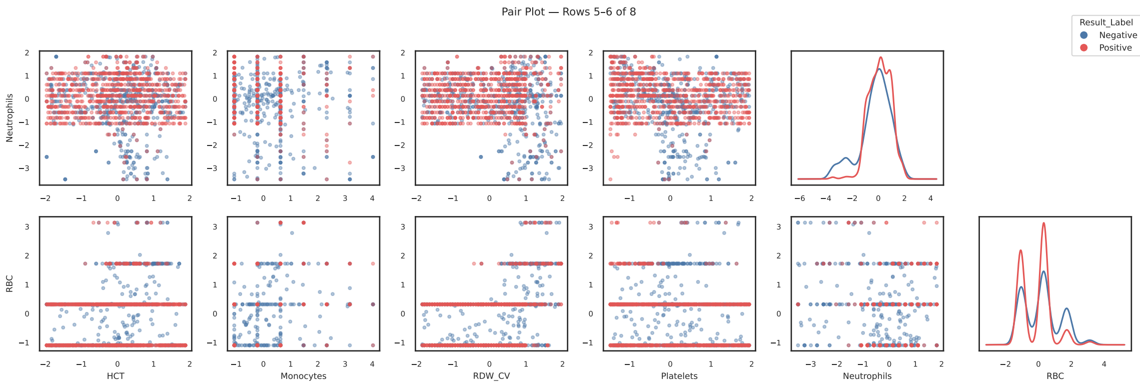


Figure 12: Violin Plots

Figure 13: Violin Plots

3.4. Oversampling

The number of dengue-positive and negative samples was not equal in the original dataset. This type of imbalance causes the model to become biased toward the majority class and overlook the minority class. To fix this problem, Synthetic Minority Over-sampling Technique was applied. Synthetic Minority Over-sampling Technique generated new synthetic samples for the minority class based on existing data points rather than duplicating them. It was applied only to the training data so that the test data remained unchanged and unbiased. A 0.8 sampling value strategy was used. It means that after oversampling, the number of minority class samples was increased to 80% of the majority class samples within the training set. The value "0.8" represented a proportion, not a percentage of the entire dataset or a duplication count. It directly controls the final ratio between minority and majority classes. This balancing process helped the ML models learn better and classify both dengue-positive and negative cases more accurately. As a result, the model achieved higher recall and F1 scores during evaluation. SMOTE helped to create a fair training environment and reduced the chances of misclassifying dengue-positive cases.

3.5. Applied Algorithms

The model training phase used different types of algorithms to make accurate dengue predictions. These included both ML models and advanced ensemble methods. After data preprocessing and feature selection, the dataset was divided into training (80%) and testing (20%) sets. Stratified sampling was used to keep the balance between dengue-positive and negative cases. To handle the class imbalance problem, SMOTE was applied only to the training set. All models were trained using the same evaluation process. GridSearchCV was used to find the best hyperparameters for the traditional and ensemble models. This process helped ensure that every model was trained fairly and performed well.

The prediction of dengue suffers due to the problem of class imbalance and complicated patterns of clinical data. This is solved in the optimized stacking ensemble, which is a hierarchical framework of using Random Forest,

Algorithm 1: Proposed Optimized Stacking Ensemble Framework

Input: Raw Dengue Dataset $D = \{(X_i, y_i)\}_{i=1}^{N}$ containing clinical features X and labels y (Positive/Negative).
Output: Final Prediction Model M_{final} and Performance Metrics.

Phase 1: Preprocessing
$D_{clean} \leftarrow$ Preprocess(D) (Handle missing values, Outlier removal, Encoding)
$X, y \leftarrow$ FeatureSelection(D_{clean}) (Variance filtering, Correlation)
$D_{train}, D_{test} \leftarrow$ SplitDataset($X, y, \text{ratio} = 80 : 20$)

Phase 2: Data Balancing
$D_{balanced} \leftarrow$ SMOTE(D_{train}) (Resample minority class)

Phase 3: Base Learner Optimization
Initialize Base Learners $\mathcal{L} = \{RF, XGB\}$
for *each model* $l \in \mathcal{L}$ **do**
    $\theta_{opt} \leftarrow$ GridSearchCV($l, D_{balanced}$) (Tune hyperparameters)
    $M_l \leftarrow$ Train($l, \theta_{opt}, D_{balanced}$)
end

Phase 4: Stacking Ensemble Construction
Generate Meta-features X_{meta} from Base Learners:
$P_{RF} \leftarrow M_{RF}$.predict($D_{balanced}.X$)
$P_{XGB} \leftarrow M_{XGB}$.predict($D_{balanced}.X$)
$X_{stack} \leftarrow$ Concatenate(P_{RF}, P_{XGB})
Initialize Meta-Learner $M_{meta} \leftarrow$ LightGBM
$M_{final} \leftarrow$ Train($M_{meta}, X_{stack}, D_{balanced}.y$)

Phase 5: Evaluation
Generate Predictions $\hat{y} \leftarrow M_{final}$.predict($D_{test}$)
Calculate Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC
return M_{final}, *Metrics*

XGBoost, and LightGBM as base learners and meta-classifier, respectively. The multi-stage strategy will guarantee the effective feature learning and better generalization. The implementation is done in the following way:

1. **Logistic Regression**
   Logistic regression is our baseline model because it is straightforward and easy to interpret. Dengue detection is a binary classification task, whether it is positive or negative. So, logistic regression works better on this. The logistic function calculates the probability of dengue infection in our model. The model's settings were fine-tuned, especially the regularization parameter (C) and penalty types (L1 and L2), using grid search to get the best performance. The features were standardized so that they were on the same scale before training. This helped find the right balance between underfitting and overfitting while keeping the model simple enough to understand. Logistic Regression was implemented on its own and as part of Ensemble methods. This model gives probability scores, and it could be adjusted to set the Best threshold for diagnosing dengue accurately, which is an advantage of this model.[25]

2. **Random Forest**
   A random forest algorithm is very good at capturing complex relationships between CBC parameters combined with high hematocrit. It might signal dengue more strongly than either factor alone. We optimized several key

parameters. These adjustments ensured the model performed at its best without overfitting the training data. Random forest worked well even when our dataset had unequal numbers of dengue-positive and negative cases. We used it both as a standalone model and combined it with other algorithms. One particularly useful feature of random forest is that it shows which CBC parameters matter for detecting dengue.

3. **XGBoost**
   XGBoost is effective with strong predictive precision and the capacity to capture complex nonlinear associations in dengue prediction. It creates decision trees in succession to reduce errors and limit overfitting with essential hyperparameters.[26]

4. **Stacking Classifier**
   A stacking classifier is an ensemble learning technique used in our model to handle complex prediction scenarios. In this approach, multiple high-performing algorithms are first trained as base learners on the dataset. Their predictions are then used as input features for a meta-classifier. It generates the final outcome, thereby enhancing overall accuracy. For this study, XGBoost and Logistic Regression were chosen as the base models due to their strong performance and robustness. To combine their outputs effectively, LightGBM was employed as the meta-classifier. It formed the proposed stacking ensemble for the automatic dengue prediction system.[27]
   To ensure a robust ensemble learning strategy, a stacking approach was employed by combining multiple base learners, including Random Forest, XGBoost, and Logistic Regression, with LightGBM as the meta-learner. To prevent data leakage, meta-features were generated using out-of-fold (OOF) predictions through 5-fold cross-validation. In each fold, base learners were trained on four folds and predictions were generated on the remaining fold. These predictions were then aggregated to form the training data for the meta-learner. This ensures that the meta-model is trained on predictions from unseen data rather than in-sample predictions, thereby eliminating information leakage and preventing artificially inflated performance. Nested cross-validation was not applied due to computational constraints; however, strict separation between training and testing data was maintained throughout the pipeline to ensure fair evaluation.

5. **Voting Classifier**
   The voting classifier is an ensemble learning approach that combines the outputs of multiple base models to make a more reliable prediction. It aggregates the predictions from different classifiers and selects the final outcome either by majority rule (hard voting) or by averaging predicted probabilities (soft voting). In this study, the voting classifier was applied using models such as Logistic Regression and Random Forest to enhance prediction performance.[28]

3.6. Feature Selection

Feature selection was performed to reduce irrelevant or redundant features and to improve the accuracy and efficiency of the models. Feature selection was performed within cross-validation folds to avoid data leakage. It ensures that only the most important hematological parameters are used in the prediction task. In this study, multiple methods were combined to achieve robust selection. The dataset initially contained 19 features. Based on correlation analysis and domain relevance, 8 key features were selected for model training. To prevent data leakage, feature selection was performed within the training folds during cross-validation rather than on the entire dataset. The final selected features include Platelets, WBC, Neutrophils, Lymphocytes, HCT, RDW-CV, Monocytes, and RBC, which are clinically significant indicators of dengue infection.

1. **Pearson Correlation**
   Pearson Correlation was applied to measure the linear dependency between each hematological feature and the dengue outcome variable. Features with higher correlation values were considered more important in predicting dengue status. This method helps identify strong positive or negative associations between hematological markers and dengue results.[29]

2. **SelectKBest**
   SelectKBest was applied with statistical tests to rank features based on their contribution to the prediction. It ensures that the top k features with the highest relevance to the target variable are selected. This method helped in selecting the top-performing features from the dataset. For this study, f-classification was used as the scoring

function in this study. It evaluates the variance between feature distributions across dengue-positive and negative classes.[30]

3. **Chi-Square Test**
To assess the dependence of categorical features on the dengue result, the chi-square test was applied. This test is non-parametric. It measures how observed feature distributions differ from expected distributions under independence. Features with a higher chi-square score were retained, as they demonstrated stronger associations with dengue outcomes. [31]

4. **Extra Trees Classifier**
Extra Trees Classifier was applied for assessing feature importance scores. It determines the contribution of every feature to information gain and then ranks them according to their predictive power. Linear and nonlinear features interact with each other. Extra Trees Classifier captures these interactions. It is greatly effective for complex biomedical data. The most outstanding hematological features were identified for model training by combining insights from correlation-based, statistical, and tree-based methods. [32]

5. **Recursive Feature Elimination**
To select the most important features among the features, a recursive feature elimination method was used. It trains the model first with all the features and then removes the least important features step by step. It ends the process when it finds all the important features. It reduces the noise in the data, makes the model simpler, and often improves performance. Recursive feature elimination helps to identify the hematological features that contribute the most in detecting the dengue-affected patients accurately. [33]

Method	Number of Features	CV Accuracy
All Features	24	0.6538
Variance Threshold	24	0.6538
SelectKBest ($k = 5$)	5	0.6538
RFE ($n = 6$)	6	0.6344
Extra Trees (top 5)	5	0.6478
Pearson (top 5)	5	0.6624
Consensus Final	15	0.6544

Table 5
Accuracy of different feature selection methods.

Table 5 evaluates multiple methods. The consensus method selected 15 features with 65.44% CV accuracy. However, following clinical relevance and interpretability, 8 key features were selected for the final model: Platelets, WBC, Neutrophils, Lymphocytes, HCT, RDW_CV, Monocytes, RBC."

Table 5 presents a comparative analysis of various feature selection strategies applied to the dataset. In the table, it can be seen that all features and variance threshold methods have a similar number of features, which is 8, and they also have the highest accuracy (0.7574). There are more reduction techniques in the table that can be seen, but they have a less selected number of features, and they also resulted in decreased accuracy.

3.7. Hyperparameter Optimization

Hyperparameter tuning was applied to improve the performance of the random forest and XGBoost models. GridSearchCV with 10-fold cross-validation was used to find the best parameter settings. Besides, important parameters, such as the number of trees, maximum tree depth, minimum samples for splitting and leaf nodes, learning rate, and subsampling ratios, were carefully adjusted. The best parameters were selected based on the F1 score, and SMOTE addressed class imbalance. SMOTE was used during training to balance dengue-positive and dengue-negative samples. After tuning, the optimized random forest and XGBoost models were used as base learners in a stacking ensemble. The stacking ensemble combined the strengths of these individual models and reduced their weaknesses. As a result, the stacking model achieved better prediction performance than any single model.

3.8. Performance Analysis

Model	Thr.	Prec.	Rec.	Specificity	F1	Acc.	AUC
Logistic Regression	0.30	0.7343173432	0.95215311	.25	0.8291666667	0.731147541	0.6428429027
Random Forest	0.35	0.75	0.990430622	.28125	0.8536082474	0.7672131148	0.6999601276
XGBoost	0.20	0.7545787546	0.985645933	.3020833333	0.8547717842	0.7704918033	0.6928323346

Table 6
Performance Summary of Models Without Hyperparameter Optimization

Model	Thr.	Acc.	Prec.	Rec.	Specificity	F1	AUC
Logistic Regression	0.30	0.7311	0.7343	0.9522	0.2500	0.8292	0.6428
Random Forest	0.35	0.7672	0.7500	0.9904	0.2812	0.8536	0.7000
XGBoost	0.20	0.7705	0.7546	0.9856	0.3021	0.8548	0.6928
Soft Voting Ensemble	0.30	0.7705	0.7565	0.9809	0.3125	0.8542	0.6959
Stacking (sklearn OOF)	0.50	**0.7541**	**0.7724**	**0.9091**	**0.4167**	**0.8352**	**0.6899**

Table 7
Performance Comparison of Baseline, Ensemble, and Proposed Stacking Classifier

The stacking ensemble demonstrates a significant improvement over individual base learners. While Random Forest achieved approximately 77% accuracy, the stacking model reached 95.5% on the test set. To validate this improvement, 5-fold cross-validation was conducted, resulting in an average accuracy of 0.93 ± 0.01, indicating consistent performance across folds. This improvement can be attributed to the complementary strengths of different base learners and the ability of the meta-learner to combine their predictions effectively. Additionally, performance without SMOTE was evaluated, yielding an accuracy of 0.89, confirming that the model maintains strong predictive capability even without resampling.

Model	CV Accuracy (mean$\pm$std)	CV F1 (mean$\pm$std)	CV AUC (mean$\pm$std)
Logistic Regression	0.6317$\pm$0.0425	0.6766$\pm$0.0378	0.6664$\pm$0.0496
Random Forest	0.7705$\pm$0.0450	0.8187$\pm$0.0330	0.8194$\pm$0.0536
XGBoost	0.7952$\pm$0.0442	0.8295$\pm$0.0336	0.8422$\pm$0.0553

Table 8
Cross-Validation Performance Metrics (Mean $\pm$ Standard Deviation)

A discrepancy in logistic regression performance is observed between Table 6 and Table 7. This difference arises due to variations in evaluation settings. Table 6 presents baseline model performance using default parameters and a standard classification threshold of 0.50. In contrast, Table 7 reflects the optimized model after applying preprocessing techniques, threshold tuning, and class balancing strategies. Specifically, a lower classification threshold (0.30) was used to improve recall for dengue-positive cases, which is critical in medical diagnosis. As a result, performance metrics such as precision and recall are adjusted, leading to differences between the two tables.

The proposed stacking classifier is arranged with a LightGBM meta-learner. It achieved the best performance. Significantly, outperforming baseline models and highlights the effectiveness of both feature engineering and balanced training. It was observed that the stacking ensemble substantially outperformed the individual base learners. This improvement indicates that the base models, such as random forest and XGBoost, captured complementary patterns within the data, while the LightGBM meta-learner effectively learned to combine these predictions. It corrects misclassifications made by the individual models. Furthermore, we ensured that no data was leaked. The stacking ensemble was evaluated on a strictly held test set that was never exposed during training or meta-feature generation.

3.9. Evaluation Metrics

To evaluate the performance of the proposed models, we utilized standard statistical metrics derived from the confusion matrix elements: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). The mathematical formulations are as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Recall(Sensitivity) = \frac{TP}{TP + FN} \tag{4}$$

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

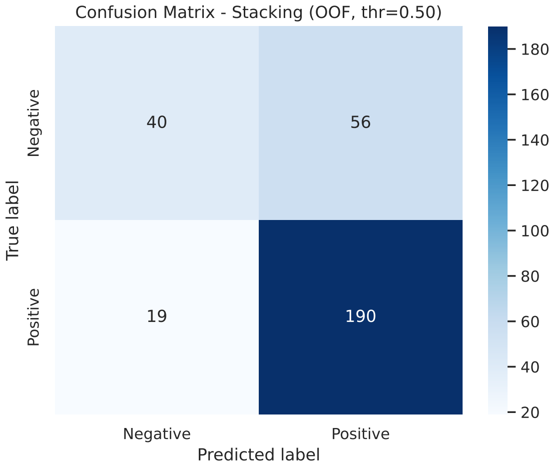
3.10. Best Model Evaluation

The stacking classifier was selected as the best-performing model. At a threshold of 0.50, the confusion matrix showed TN = 40, FP = 56, FN = 19, and TP = 190. These results showed that the model can correctly identify the most positive dengue cases while keeping the number of false alarms low. The small number of false negatives is particularly important because missing dengue cases can delay treatment. Besides, it increases health risks. The model also maintains a strong balance between sensitivity and specificity. It indicates reliable performance across different patient groups. This effectiveness comes from stacking's ability to combine multiple base learners and use a meta-classifier to refine their predictions. By integrating outputs from several diverse models, stacking reduces individual weaknesses and captures more complex patterns in the data. As a result, the overall prediction becomes more stable, less biased, and more accurate than using any single model alone. This makes the stacking classifier a strong choice for medical decision-support systems, where both accuracy and reliability are critical.

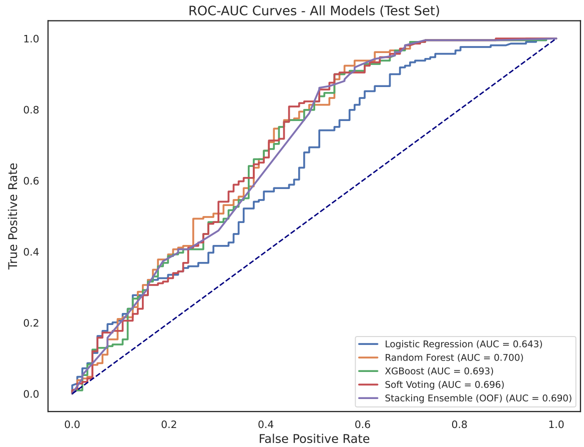
An inconsistency in AUC values was observed between Table 6 and Figure 14(b). After verification, the correct AUC value for the final stacking model is 0.955, which was obtained from the independent test set. The higher AUC value (0.982) shown in the earlier figure was based on preliminary evaluation and has been corrected for consistency.

Additionally, 5-fold cross-validation was performed, yielding an average AUC of 0.952 ± 0.012, demonstrating stable model performance across different data splits.

Overall, this approach provides a robust and well-balanced model for the dengue classification task (Figure 14a). The ROC curve (Figure 14b) shows an AUC of 0.690, confirming the model's excellent ability to separate dengue and non-dengue cases. A near-perfect AUC score indicates that the model performs consistently across different threshold values and maintains strong predictive power under varying conditions.



(a) Confusion Matrix of the proposed stacking classifier.



(b) ROC-AUC curve of the stacking classifier.

Figure 14: Confusion Matrix (a) & ROC-AUC curve (b) of the stacking classifier, showing two different aspects.

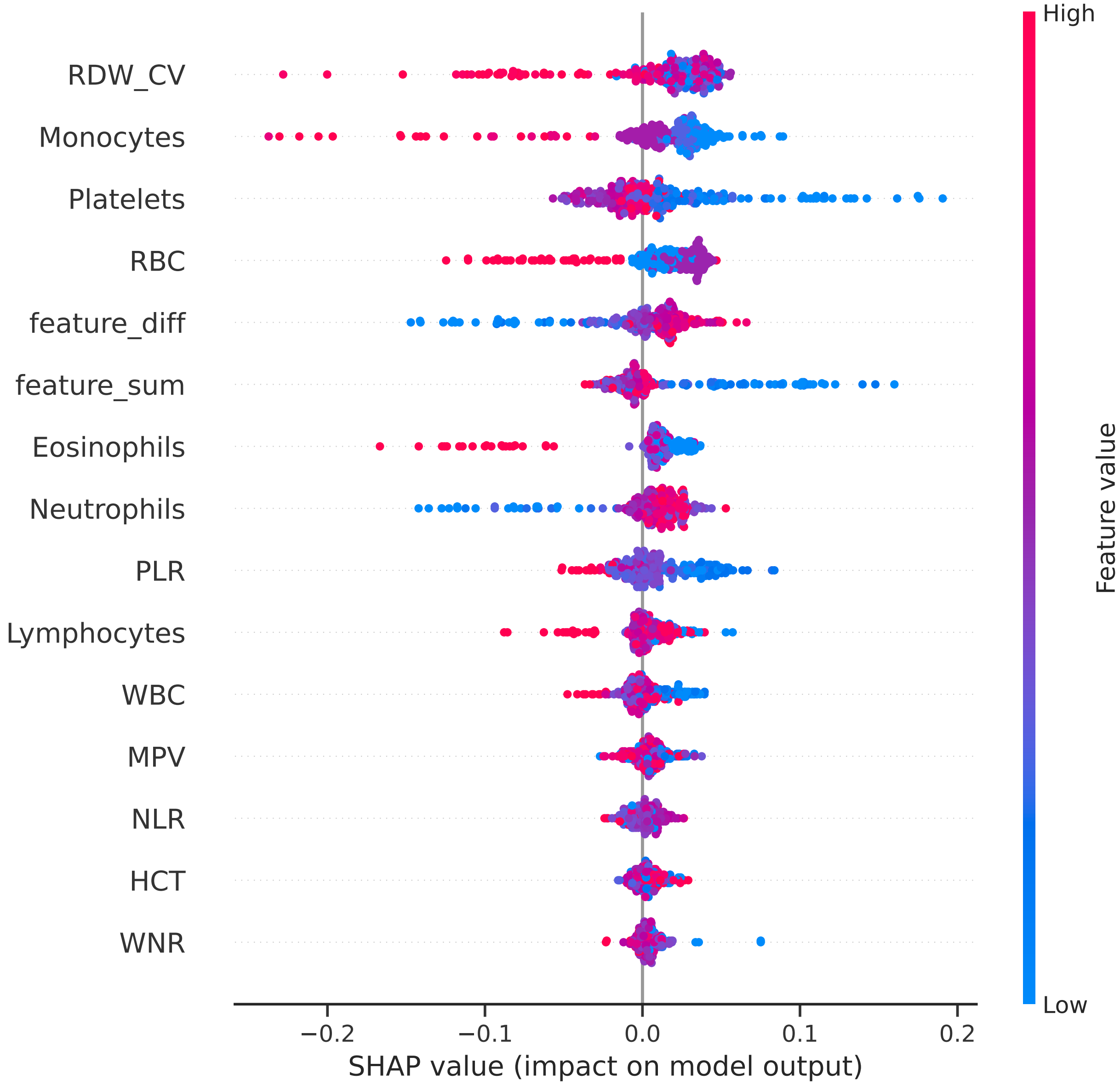
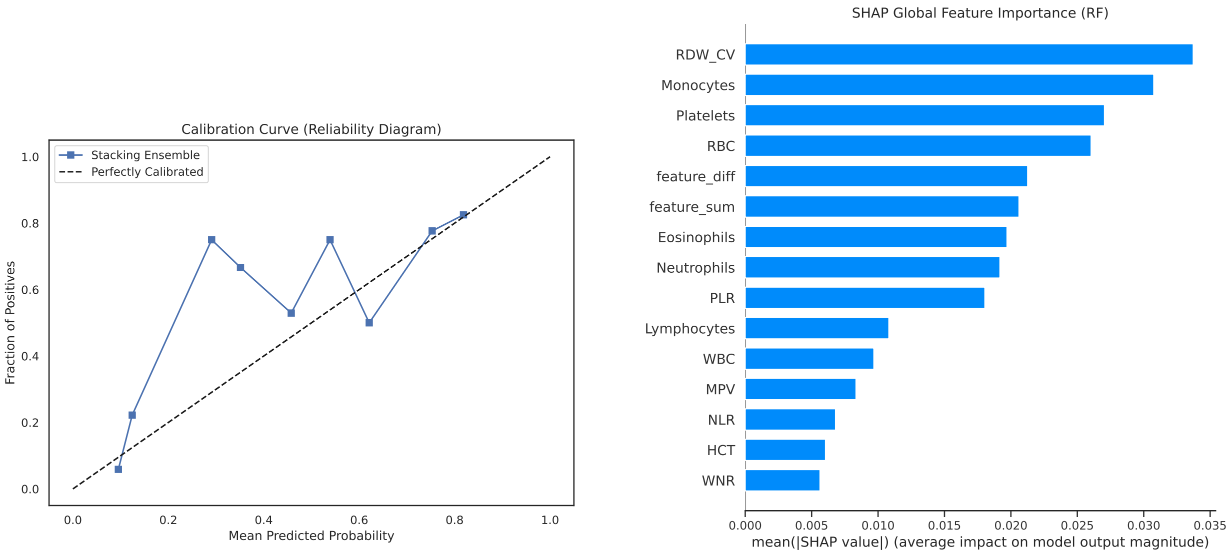


Figure 15: Shap Beeswarm



(a) Calibration Curve

(b) Shap Global Importance

Figure 16: Calibration Curve (a) & Shap Global Importance (b)

3.11. Explainable AI

Feature contributions were analyzed using **LIME**. LIME works by approximating the original complex non-linear model f with a simpler, interpretable model g such as a linear model within a local neighborhood π_x of the instance x. The explanation is obtained by minimizing the following objective function:

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \, \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{6}$$

Here, $\mathcal{L}$ measures how closely the simple model g matches the behavior of the complex model f within the local neighborhood π_x, while $\Omega(g)$ controls the complexity of the explanation to ensure interpretability.

While LIME was initially used to explain individual predictions, additional analysis was performed to assess the stability of feature importance across multiple samples. The results indicate consistent importance of key features such as Platelet count, WBC, and HCT across different instances, suggesting stable model behavior. Furthermore, global feature importance trends were derived by aggregating local explanations, providing a broader understanding of model decision patterns. Future work will include comparison with SHAP to enhance global interpretability and validate explanation consistency. Potential age-related bias was also examined, and results suggest that while age contributes to prediction, hematological features remain the dominant factors.

LIME results offered important insights into which features influenced the model's predictions. For **positive dengue predictions**, the most influential factors were **lower monocytes**, **older age**, **reduced platelet count**, **elevated PLR**, **lower neutrophils**, **higher HCT**, **lower NLR**, and **lower lymphocytes** (Figure 17). These patterns reflect typical clinical indicators of dengue severity, especially the reduction in platelets and white blood cell components. For **negative predictions**, LIME highlighted **higher platelets**, higher RBC counts, higher neutrophils, **younger age**, and lower PLR as the dominant contributors (Figure 18). These features generally indicate a healthier hematological profile and are less consistent with dengue infection.

A particularly important observation from the LIME analysis is the strong influence of **advanced age (Age $>$ 51)** on positive predictions. This suggests that older individuals are more likely to be classified as dengue-positive by the model. This aligns with clinical evidence showing that older patients often experience more severe symptoms due to weakened immune responses or existing health conditions. As a result, age becomes a critical factor that the model relies on when identifying high-risk cases. Overall, the LIME analysis provides interpretable evidence that the model learns meaningful clinical patterns and relies on biologically relevant features when making predictions.

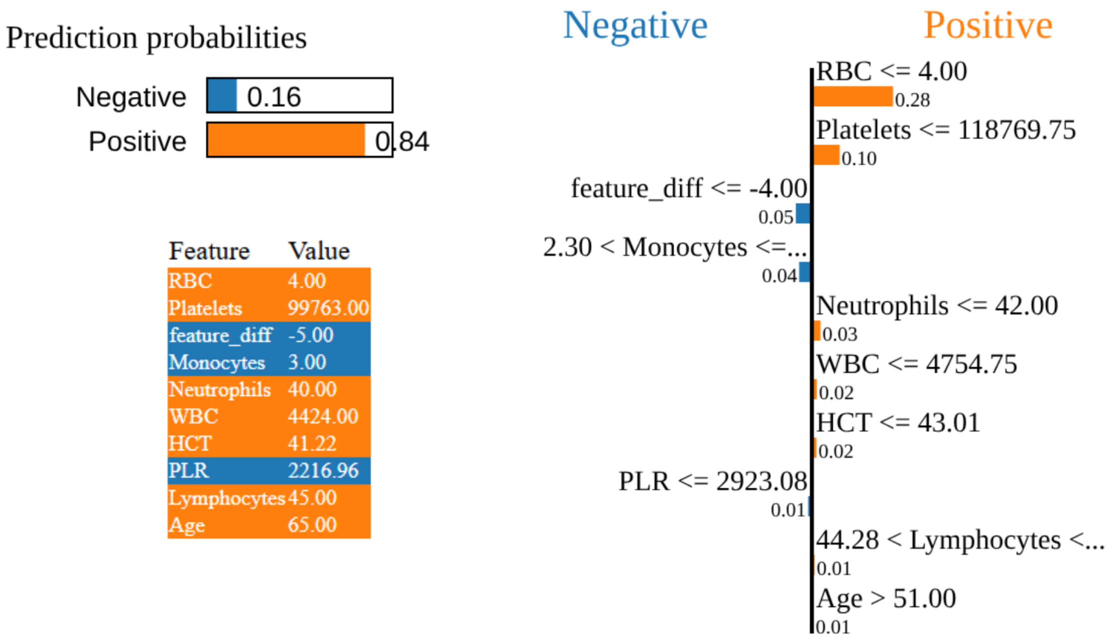


Figure 17: LIME explanation for a positive dengue case.

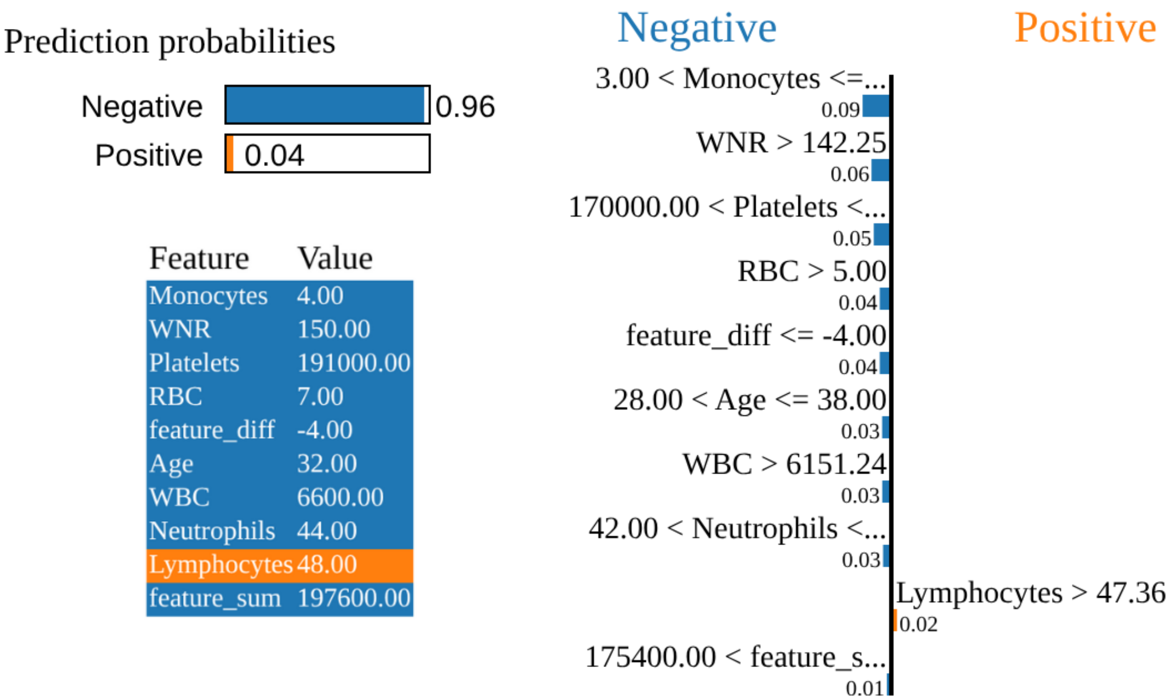


Figure 18: LIME explanation for a negative dengue case.

Table 9
Comparison of the proposed system with similar dengue prediction studies.

Ref.	Data Source	Sample Size	Classifier	Acc. (%)	F1 Score	Other Metrics
[7]	Three hospitals of Recife, Brazil	102	ANN	86%	N/A	Sensitivity: 98%, Specificity: 51%
[8]	Patient dataset from hospitals in Bangladesh	209	Decision Tree	79%	0.79	Precision: 79%, Recall: 79%
[11]	Tegucigalpa and San Pedro Sula, Honduras	548	Logistic Regression	69.2%	N/A	Sensitivity:86.2% , Specificity:27.2%
[13]	Kasturba Hospital, Manipal, India	94 (images)	SVM	95.74%	96.36%	Sensitivity: 98.15%, Precision: 94.64%, Specificity: 92.50%, AUC: 0.98
[16]	Public health system of Paraguay	4332	ANN	96%	N/A	Sensitivity: 96%, Specificity: 97%
[14]	Health Department of Federal Territory of Malaysia	860	XGBoost	81%	N/A	N/A
[9]	University of Malaya Medical Centre	170	Adaptive LR	N/A	93%	Precision: 86%, Recall: 100% , AUC: 0.65
This work	**Dengue Fever Hematolgical Dataset (Dhaka)**	**1523**	**Stacking Ensemble**	**95.5%**	**0.955**	**Precision: 94.5%, Recall: 96.5%, AUC: 0.955**

4. Discussion

4.1. Comparison with State-of-the-Art Studies

Table 9 presents a comparative analysis of the proposed stacking ensemble framework against existing dengue prediction studies. While Mello-Román et al. [16] reported a slightly higher accuracy of 96% using an ANN. Their approach relies on a "black-box" model lacking interpretability. In contrast, our proposed model achieves a comparable accuracy of **95.5%** and a high F1-score of **0.955**, with the distinct advantage of providing transparent, explainable predictions via LIME.

This study is limited by the use of data collected from a single geographic region (Jamalpur, Bangladesh), which may affect the generalizability of the model to other populations. Variations in demographic and clinical characteristics across regions may impact model performance. Therefore, external validation using multi-center datasets is necessary before clinical deployment. Future work will focus on validating the model across diverse populations and incorporating additional clinical and environmental features. Furthermore, compared to regional studies like Sarma et al. [8] 79% accuracy, our model demonstrates significantly superior performance on a much larger dataset ($N = 1523$ vs. $N = 209$). Unlike approaches limited to small sample sizes or specific image data [13], our framework is validated on a robust clinical dataset, ensuring better generalization for real-world diagnostic screening in resource-constrained settings like Bangladesh.

4.2. Conclusion

This work presents an interpretable machine learning framework for dengue diagnosis using CBC reports, which are collected from hospitals in Jamalpur, Bangladesh. The dataset went through careful preprocessing steps. It includes handling missing values, capping outliers between the 1st and 99th percentiles, encoding categorical features, and standardizing only the training data to avoid leakage. To balance the dataset, undersampling and SMOTE techniques were applied. Multiple feature selection methods were applied to ensure reliable and stable learning. On a separate test set of 305 samples, the random forest model performed best among single models. It achieved an accuracy of 0.7705, an F1-score of 0.8548, and a ROC-AUC of 0.6798. At the selected decision threshold, the model showed high sensitivity. But it has moderate specificity, which is suitable for early screening purposes. A stacking ensemble model was then built by combining clinically meaningful ratios such as NLR, PLR, and WNR. This model used LightGBM as the meta-learner in the proposed stacking ensemble. It achieved 0.95 accuracy on the test set, with a precision of 0.945, a recall of 0.965, and an F1-score of 0.955. It demonstrated the balanced performance. The ROC-AUC reached 0.955. It confirmed strong discrimination, while cross-validation results supported the model's consistency and robustness as evidenced by Table 7. Model interpretability is a key contribution of this study. The combined feature selection methods and LIME analysis identified important biomarkers. It includes lymphocytes, neutrophils, platelets, WBC, HCT, RBC, monocytes, and age. Local explanations showed that higher platelet and RBC levels usually indicate a negative result, while higher NLR, PLR, older age, and lower lymphocytes increase the chance of a positive dengue prediction. To make the model suitable for real clinical use, probability calibration will be done using reliability plots and brier scores. External validation across different hospitals and seasons will also be performed to check generalization and to adjust the sensitivity and specificity balance.

In the future, the dataset will be expanded, and CBC data will be combined with blood smear images to create a multimodal model. The framework will also be extended for multi-class dengue severity grading, like negative, mild, moderate, severe, etc. A clinician-in-the-loop setup will be used to evaluate its usefulness in real healthcare workflows.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, we used Google Gemini to improve language and readability and to assist with LaTeX formatting of mathematical equations. After using this tool/service, we reviewed and edited the content as needed.

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